Game Theory-Based Hyperspectral Anomaly Detection

Zhihong Huang, *Student Member, IEEE*, Xudong Kang, *Senior Member, IEEE*, Shutao Li, *Fellow, IEEE*, and Qiaobo Hao, *Student Member, IEEE*

***Abstract*— In this article, a novel game theory-based approach is proposed for anomaly detection in hyperspectral images (HSIs) via effectively exploring multiple spectral and spatial characteristics of anomalies. This approach comprises three main steps. First, spectral, extend morphological profiles (EMPs), and Gabor texture features are captured from an input HSI. Then, we define the anomaly detection problem as an anomaly game model, in which image regions (superpixels) of different features are modeled as players who select to be "anomaly" or "background" as their strategies. Three initial detection results are produced based on each player's strategy in the Nash equilibrium of the anomaly game. Last, a saliency-based decision fusion technique is used to combine the complementary information in different features, so as to obtain a fused detection map. The performance of the proposed anomaly detection technique is evaluated on four real-scene HSIs. Experimental results validate that our approach can outperform some state-of-the-art anomaly detection methods.**

***Index Terms*— Anomaly detection, decision fusion, game theory, hyperspectral images (HSIs), Nash equilibrium, spectral–spatial information.**

I. Introduction

HYPERSPECTRAL anomaly detection has attracted much interest in recent years [1]–[4], since it plays a significant role in remote sensing applications, such as military reconnaissance [5], [6], mineral exploration [7], [8], and environmental monitoring [9]. It is an unsupervised technique for automatic target detection [10]–[12], which aims to discriminate anomaly targets from their surrounding background [13]–[15].

Anomalies usually refer to pixels that have distinct spectral–spatial difference with their surroundings [16]–[19]. In general, they have two anomalous characteristics, spectral anomaly and spatial anomaly. Spectral anomaly means that anomaly and background pixels have a difference in spectral signal [20]. Different coverings usually reflect various spectral signals [21]. Spatial anomaly represents that anomaly and background pixels exhibit a difference in spatial distribution. Anomalies usually are small targets with low probabilities of appearance [22], [23]. For example, warships in a sea background and trucks in a highway background can be considered as anomaly objects.

According to the characteristics of anomalies, the existing anomaly detectors fall into two major types: statistics modeling techniques [24]–[35] and representation-based techniques [36]–[45]. Statistics modeling techniques are usually associated with Gaussian distribution. The Reed–Xiaoli detector (RX) [24] is a well-known statistics modeling method. It characterizes the background by utilizing a multivariate Gaussian model. Then, anomalies can be identified by calculating the Mahalanobis distance between test pixels and background pixels. Moreover, several improved RX-based detection techniques have been developed [25]–[30]. The local RX (LRX) detection method [25] adopts local statistics to estimate background pixels. The regularized-RX detection method [26] standardizes a covariance matrix of images to improve detection performance. The kernel-RX detection method [28] models hyperspectral images (HSIs) into high-dimension feature spaces that can strengthen differences between target and background classes. Besides the above Gaussian model-based methods, some robust statistics modeling techniques [31]–[35] have been designed including probabilistic anomaly detector [31], linear mixing model [32], and discriminative metric learning-based anomaly detector [33].

Moreover, representation-based methods have also drawn much attention [36]–[39]. These methods assume that background pixels can be represented by a set of main bases, whereas anomaly pixels cannot [40]–[42]. In [36], the background joint sparse representation detector (BJSRD) employs an orthogonal subspace to obtain background bases and then detects anomalies. The collaborative representation-based detector (CRD) [40] supposes that targets can be identified from a residual image that is produced via subtracting a predicted image from an input hyperspectral data set. The LRaSMD-based Mahalanobis-distance anomaly detection (LSMAD) approach [43] uses low-rank representation to decompose low-rank background and sparse anomaly parts.

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In general, most previous methods tackle the anomaly detection problem by minimizing one single energy function. Due to the complexity of real scene HSIs, one single energy function might not characterize the difference between anomaly pixels and background pixels effectively. To tackle this issue, in this article, a novel game theory-based detection (GTD) approach is proposed for detecting anomalies in HSIs. In the GTD method, one specific payoff function is defined for each superpixel. This payoff function comprises multiple energy functions, which can jointly utilize the spectral–spatial characteristics of anomalies and affinities among superpixels. Using multiple energy functions can make our detector more robust in a complex background. The reason is that when one is unsatisfactory, the others may be effective. Specifically, this approach comprises three main steps. First, spectral, extend morphological profiles (EMPs), and Gabor texture features are extracted from an input HSI. Then, we define the anomaly detection problem as an anomaly game model, where image regions (superpixels) of different features are modeled as players who select to be "anomaly" or "background" as their strategies. Three initial detection results are produced based on each player's strategy in the Nash equilibrium of the anomaly game. Last, we exploit the complementary information in three features with a saliency-based decision fusion technique, so as to obtain a fused detection map. Compared with some state-of-the-art detection methods, experimental results demonstrate that our method can achieve a competitive detection performance on four real hyperspectral data sets. The major contributions of this article are emphasized in the following.

1) With a new strategic interaction perspective, anomaly detection has been modeled as an anomaly game problem. The spectral–spatial characteristics of anomalies are used to construct the payoff function in the anomaly game. To our knowledge, the game theory is applied in hyperspectral anomaly detection for the first time.
2) The complementary information of different spectral and spatial features are combined with a novel decision fusion technique. It is demonstrated that the fusion of spectral and spatial features could lead to an obvious improvement in detection performance.

The rest of this article is summarized as follows. In Section II, the related works in game theory are reviewed and some notions in this article are introduced. Section III introduces the proposed GTD method. Section IV gives the experimental results. Finally, the conclusion of this article is presented in Section V.

II. Related Works and Definitions

In this section, we first review the related works in game theory. Then, some notions in this article are introduced that will be used throughout this article.

A. Game Theory

Game theory studies strategic interaction between decision-makers through mathematical models, and it aims to find an optimal decision scheme under the condition of conflict confrontation. This theory has been widely developed in social science, logical, and computer science fields since the 1940s.

Recently, game theory has been successfully employed in machine learning and computer vision [46]–[51]. In [46], a clustering game in data points is developed based on a game-theoretic perspective. In [47], a noncooperative game is proposed to match two objects. In [48], a game-theoretic integration method is proposed for image segmentation. A feature subspace selection method [49] is developed by using the principle of coalition game theory. A multiplayers game [50] is modeled for transduction learning. Zeng *et al.* [51] find that the saliency detection problem can be modeled as a game process by estimating the Nash equilibrium. It should be noted that the Nash equilibrium is a common equilibrium state in which no player can improve the payoff by the change of strategy when its opponents no longer change their strategies. Inspired by this creative idea, in this article, the anomaly detection problem is modeled as an anomaly game model, and the Nash equilibrium is the key tool to detect the anomalies [52].

B. Notions

1) Player: All participants involved in the game are referred to as players, in which each player can change his strategy independently to maximize the payoff. Let i represent the ith player. In our method, one superpixel is used to represent one player.

2) Strategy: The competition scheme that each player could choose is defined as a strategy. z_i is used to represent the strategies of the ith player. In this article, to be "background" and "anomaly" are the two different strategies for each player.

3) Payoff: The result of the game is that the players can obtain different payoffs by adopting different strategies. The payoffs are dependent on all players' strategies. Let u_i describe the ith player's payoff. We jointly exploit spectral and spatial anomaly properties to construct a payoff function in our anomaly game. More details will be introduced in Section III.

III. Proposed Method

Fig. 1 displays the schematic of the GTD approach which consists of three main steps: spectral–spatial feature extraction, anomaly game, and saliency-based decision fusion. First, spectral, EMPs, and Gabor texture features are extracted from an input HSI. Then, the anomaly detection problem is formulated as an anomaly game model, in which image regions (superpixels) of different features are modeled as the players who select to be "anomaly" or "background" as their strategies. Three initial detection results are produced by estimating the Nash equilibrium of the anomaly game. Last, the initial detection results are combined with a saliency-based decision fusion technique so as to obtain a fused detection map.

A. Spectral-Spatial Feature Extraction

The main information of HSIs lies on two spatial dimensions and one spectral dimension. In this article, we first extract the spectral–spatial features (F_n) from the HSIs, in which F_1, F_2, and F_3 represent the spectral, EMPs, and Gabor texture features, respectively. The description of these features is as follows.

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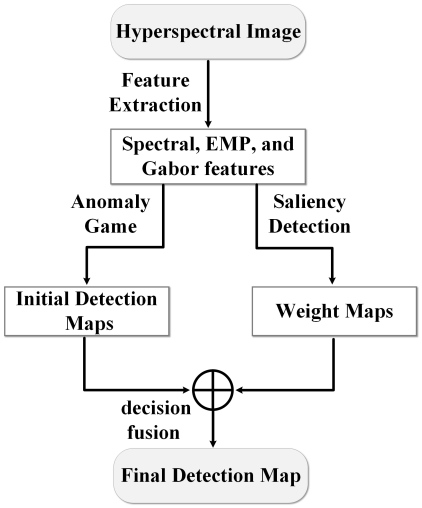


Fig. 1. Schematic of the proposed GTD approach.

1) Spectral Feature F_1: This feature refers to a dimension reduced HSI data. A simple average fusion method is adopted to generate the dimension reduced HSI data. Since the adjacent bands have a high correlation, the fused image can effectively remove the redundant information and preserve main structures for each subgroup. Specifically, the HSI is first divided into K subgroups of hyperspectral bands. Then, the bands in each subgroup are fused by averaging. The resulting image of K bands is the dimension reduced HSI.

2) EMP Feature F_2: A morphological profile (MP) can be produced by using a series of morphological opening and closing operations with different size of structuring elements (si) on the first three principal components of HSIs. The shape of the structuring element is circular. When multiple MPs are combined, the EMP feature can be generated. EMP feature can characterize the morphological structure information in HSIs. The number of opening/closing is represented as c.

3) Gabor Texture Feature F_3: This feature is obtained by employing a series of Gabor filters on the first principal component of HSIs. The Gabor filters can exploit the spatial texture information in various scales and directions [6], which are constructed by multiplying a Gaussian function with a sine wave. We use λ and ϕ to represent the wavelength and rotation angle in the sine wave.

Note that, since these features characterize the HSIs from various aspects (spectrum, morphology, and spatial texture), it is natural to utilize their properties for detection jointly. In this article, a set of default parameters $K = 20, c = 3$, $si \in (1, 2)$, $\lambda \in (1, 2, 3)$, $\phi \in (30°, 60°, \ldots, 180°)$ are adopted in the experiments. Moreover, the dimensions of features F_1, F_2, and F_3 are set as 20, 21, and 18 for four test data sets, respectively.

B. Anomaly Game

1) Game Setting: In the anomaly game, superpixels in three features are used to represent the players. Superpixels are obtained with the entropy rate superpixel (ERS) [53] algorithm, which has been demonstrated to be a quite useful tool in the field of HSI classification [6]. Based on the first three principal components of the input HSI, the ERS can produce a lot of edges by solving a graph optimization problem [53]. $I = (1, 2, \ldots, N)$ is a group of superpixels, and N is the number of superpixels. The ith superpixel is considered as the ith player ($i \in I$).

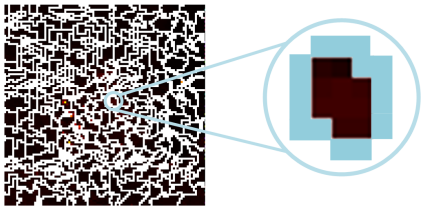


Fig. 2. Detection region $ano^1(i)$.

We assume that each player (superpixel) has two strategies, to be "background" and "anomaly." z_i^0 represents the probability that selects to be "background", and z_i^1 means the probability that selects to be "anomaly" ($z_i^0 + z_i^1 = 1$). Probability distributions in the two strategies are defined as a mixed strategy $z_i = (z_i^0, z_i^1)^T$. We also define $Z = (z_1, z_2, \ldots, z_N)$ as a set of mixed strategies for all players.

2) Payoff Construction: When the ith player uses its mixed strategy z_i against the jth player's mixed strategy z_j, its payoff is termed as $\mu_{ij}(z_i, z_j)$. The payoff $\mu_{ij}(z_i, z_j)$ of the ith player for its two-person game is modeled as follows:

$$\mu_{ij}(z_i, z_j) = \lambda_1 ano_i^1 + \lambda_2 ano_i^2 + \sup_{ij}(z_i, z_j) \tag{1}$$

where ano_i^1 and ano_i^2 are the ith player's two anomaly terms. $\sup_{ij}(z_i, z_j)$ is the ith player's support term that the jth player provides to the ith player. The ano_i^1 term can be calculated by using the LSMAD method [43] as follows:

$$ano_i^1 = \frac{1}{N_i^1} \sum_{n=1}^{N_i^1} v_n^1 \tag{2}$$

where v_n^1 means the intensity value of the nth pixel in $ano^1(i)$. $ano^1(i)$ is the ith superpixel's detection region in a detection result produced by the LSMAD method. Fig. 2 illustrates the detection region $ano^1(i)$. N_i^1 is the number of pixels included in $ano^1(i)$. ano_i^1 is the average value of all constituent pixels in $ano^1(i)$. The ano_i^2 term can be calculated by using the RX method [24] as follows:

$$ano_i^2 = \frac{1}{N_i^2} \sum_{n=1}^{N_i^2} v_n^2 \tag{3}$$

where v_n^2 is the intensity value of the nth pixel in $ano^2(i)$. $ano^2(i)$ means the ith superpixel's detection region in a detection result produced by the RX method. N_i^2 is the number of pixels included in $ano^2(i)$. ano_i^2 is the average value of all constituent pixels in $ano^2(i)$. λ_1 and λ_2 are the weight parameters that control the contributions of the first two terms. The RX and LSMAD methods are typical statistics modeling and representation-based methods, respectively. They both have the competitive performance on detection accuracy and computation cost. Their effectiveness and efficiency have

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been demonstrated in the related publications [11], [13], [18]. Using two anomaly terms makes our detector more robust. The reason is that when one is unsatisfactory, the other may be effective. Two parameters are empirically set to be $\lambda_1 = 1 \times 10^{-7}$, $\lambda_2 = 1 \times 10^{-7}$. For the support term, when playing with the opponent, each player measures the opponent's strategy. Specifically, if the opponent's strategy is similar to its strategy, the ith player will give great support to its opponent. Otherwise, the ith player will provide little support to its opponent. We define the support term as follows:

$$\sup_{ij}(z_i, z_j) = \begin{cases} A_n(i,j) - \dfrac{\beta}{N}\displaystyle\sum_{k=1}^{N} A_n(i,k), & \text{if } z_i = z_j \\ 0, & \text{if } z_i \neq z_j \end{cases} \tag{4}$$

where β represents a constant andis empirically set to $\beta = 7 \times 10^{-3}$. $A_n(i, j)$ means the affinity between the ith player and the jth player in the nth feature.

The affinity between the ith player and jth player in the spectral feature is termed as $A_1(i, j)$. $A_1(i, j)$ can be modeled by using a Gaussian weighted distance

$$A_1(i,j) = e^{-\sigma \| f_i^1 - f_j^1 \|} \tag{5}$$

where f_i^1 represents the spectral feature vector of the ith superpixel. σ is a Gaussian weight parameter. Similarly, $A_2(i, j)$ and $A_3(i, j)$ are affinities in the EMP and Gabor texture features space, respectively.

We define the expected payoff $\mu_i(Z)$ that the ith player acquires via competing with others based on a mixed strategy game as follows:

$$\mu_i(Z) = \sum_{j \neq i} \mu_{ij}(z_i, z_j). \tag{6}$$

3) Estimating Equilibrium: To estimate the Nash equilibrium in the anomaly game, a replicator dynamics (RD) method [54] is utilized, iterating until $|z_i(t) - z_i(t-1)| < \epsilon$

$$z_i(t+1) = z_i(t) \frac{c + \sum_{j \neq i} \mu_{ij}(e^h, z_j)}{c + \sum_{j \neq i} \mu_{ij}(z_i, z_j)} \tag{7}$$

where $z_i^h(t)$ is the hth constituent of the ith player's mixed strategy at time t. e^h represents a vector where the hth constituent is 1, and others are 0. c is a positive number. The ith player's initial mixed strategy is empirically set as (0.5, 0.5). When the game reaches the Nash equilibrium, each player uses its strategy to maximize its payoff if other players' strategies are unchanging. An initial detection result D_n can be generated based on each player's mixed strategy in the Nash equilibrium of our anomaly game. In our anomaly game, each player observes the opponents' strategies and provides support. Whatever strategy each player uses, it has supporters and protesters. Game equilibrium can offer a satisfactory balance in various influences. With the designed payoff in anomaly game, each player can employ a strategy which suits itself best based on its anomaly property and supports from others. Moreover, the detailed procedure of the anomaly game step is illustrated in Algorithm 1.

Algorithm 1 Pseudocode of Anomaly Game

Input: Input feature F, and the number of superpixel N.
Output: The probability that ith player selects to be "anomaly" or "background" z_i.
1: Segment F into N homogenerous superpixels by using the ERS method [53];
2: Construct the payoff $\mu_{ij}(z_i, z_j)$ of the ith player by using the RX [24] and LSMAD [43] methods;
3: Estimate the Nash equilibrium in anomaly game and obtain z_i via employing the RD method [54];
4: **return** z_i.



(a) (b)

Fig. 3. San Diego data set. (a) Pseudo-color image. (b) Reference detection map.

C. Saliency-Based Decision Fusion

In this step, the initial detection maps obtained with different features are fused with a weighted averaging method

$$\hat{D} = \sum_{n=1}^{3} D_n \times W_n \tag{8}$$

where $\hat{D}$ is the final detection map. D_n is the nth initial detection map obtained by the nth feature, and W_n is the nth weight map, which is estimated by a saliency detection method (spectral residual (SR) model) [55]

$$W_n = \mathcal{F}^{-1}(\mathcal{L}(F_n) - B_n) \tag{9}$$

where $\mathcal{F}^{-1}$ is the inverse Fourier transform. $\mathcal{L}(F_n)$ is the log spectrum of the nth feature in the Fourier domain. B_n is the averaged spectrum which is calculated as follows:

$$B_n = h * \mathcal{L}(F_n) \tag{10}$$

where h is an $m \times m$ matrix defined by

$$h = \frac{1}{m^2} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \tag{11}$$

where m equals 3.

IV. Experiments and Discussion

A. Data Sets

Four real HSIs are used to investigate the detection performance of the proposed method. They can be acquired from a public website[1]. These data sets are described as follows.

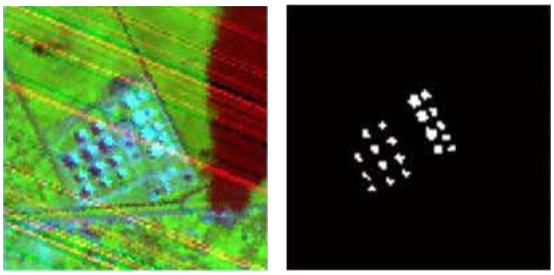
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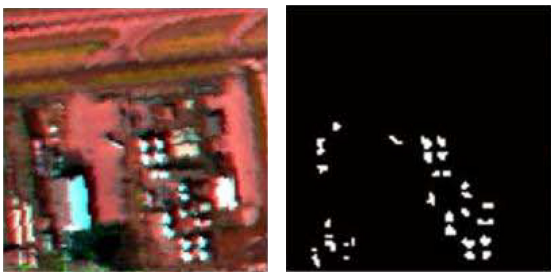
(a) (b)

Fig. 4. Pavia City data set. (a) Pseudo-color image. (b) Reference detection map.



(a) (b)

Fig. 5. Texas Coast data set. (a) Pseudo-color image. (b) Reference detection map.



(a) (b)

Fig. 6. Los Angeles data set. (a) Pseudo-color image. (b) Reference detection map.

1) San Diego Image: It is widely employed in related publications. This image was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor. It is a scene of San Diego airport area, which contains 100×100 spatial pixels and 189 spectral channels spanning the wavelength range of 0.370–2.51 μm. The spatial resolution of this image is 3.5 m/pixel. Three airplanes in this data set are regarded as anomalies. The background includes hangars, parking aprons, and exposed soil. Some background areas (such as the hangar) have similar materials with airplanes. This may result in difficulties in identifying targets from their surroundings. Anomaly targets consist of 134 pixels and account for 1.34% of the image scene. This data set and reference map are shown in Fig. 3.

2) Pavia City Image: This image was captured by the reflective optics system imaging spectrometer (ROSIS) sensor. It is a scene of Pavia city area. It has 150×150 pixels and the spatial resolution is 1.3 m/pixel. It comprises 205 spectral channels from 0.43 to 0.86 μm. The kinds of background are various, including bridge, river, bare soil, and buildings. We consider vehicles on the bridge as anomalies. Anomaly targets consist of 68 pixels and account for 0.30% of the image scene. This data set and reference map are presented in Fig. 4.

3) Texas Coast Image: It was captured by the AVIRIS sensor. It is a scene of the Texas coast area. It has 100×100 pixels, and 207 spectral channels spanning the wavelength range of 0.45–1.35 μm. Its spatial resolution is 17.2 m/pixel. It is a challenging data set due to the corruption of the serious strip noise. Buildings in this image are regarded as anomaly targets. Anomaly targets consist of 155 pixels and account for 1.55% of the image scene. The test data set and reference map are shown in Fig. 5.

4) Los Angeles Image: It was obtained by the AVIRIS sensor. It is a scene of Los Angeles city area. It has 100×100 pixels and 205 spectral channels from 0.43 to 0.86 nm. The spatial resolution of this data set is 7.5 m. In this data set, we consider houses as the objects to be detected. Anomaly targets consist of 212 pixels and account for 2.12% of the image scene. The test data set and reference map are presented in Fig. 6.

B. Experimental Setup

Five state-of-the-art detectors are employed for comparison, including the RX [24], LRX [24], CRD [40], BJSRD [36], and LSMAD methods [43]. The RX approach is a well-known detector in the unsupervised detection field. This detector identifies targets by estimating the global background statistics. The LRX method is an improvement version of the RX method, which detects objects by estimating the local background statistics. The RX and LRX methods are representative statistics modeling approaches. The CRD, BJSRD, and LSMAD detectors are classical representation based anomaly detection approaches.

To demonstrate the effectiveness of the proposed detector, the receiver operating characteristic (ROC) curve is employed, which is a widely used evaluation index in target detection [57], [58]. It is estimated via plotting the detection probability (DP) and the false alarm rate (FAR) at various thresholds [59], [5]. Moreover, we also use the area under the curve (AUC) value to evaluate the detection results. This estimates the area of the ROC curve that provides numerical accuracy measures [60]–[62].

The inner window size in the LRX and BJSRD approaches w_{in} is selected optimally from 3 to 19, and the outer window size w_{out} is selected optimally from 5 to 23. The rank r in the LSMAD approach is set to 1. The regularization parameter λ in the CRD approach is fixed as 10^{-6}.

C. Detection Performance

For the San Diego data set, the detection results of different methods are illustrated in Fig. 7. It can be observed that the detection map produced by the GTD method can detect three airplanes with high intensities. The LRX, BJSRD, and LSMAD detectors can identify anomalies, but their intensities are lower than the GTD method. The RX and CRD methods

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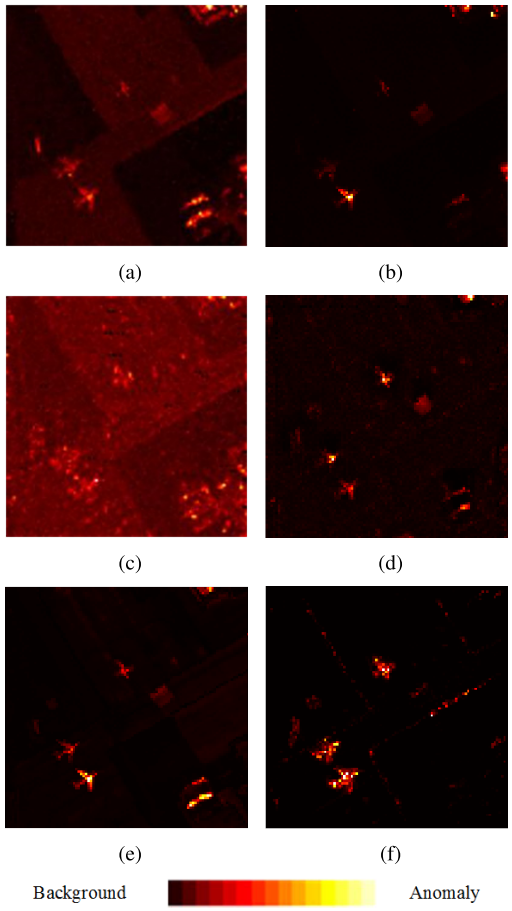


Fig. 7. Detection maps obtained by different methods for the San Diego data set. (a) RX (AUC = 0.9403). (b) LRX (AUC = 0.9457). (c) CRD (AUC = 0.9514). (d) BJSRD (AUC = 0.9643). (e) LSMAD (AUC = 0.9788). (f) GTD (AUC = 0.9882).

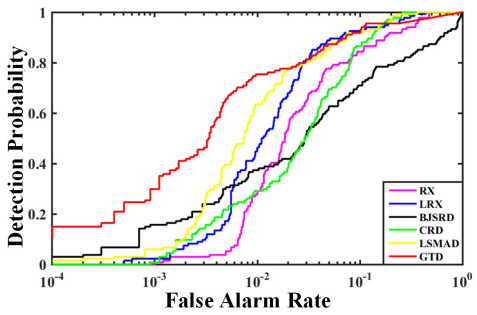


Fig. 8. ROC curves for the San Diego data set.

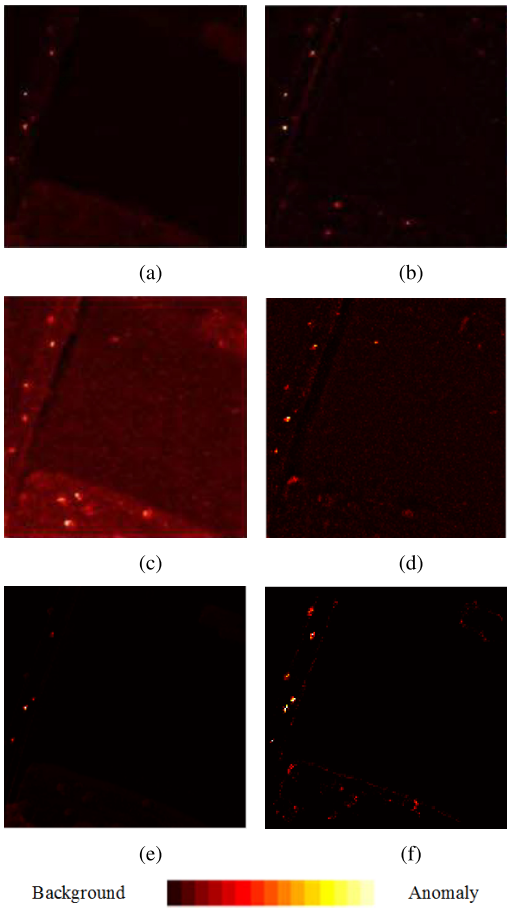


Fig. 9. Detection maps obtained by different methods for the Pavia City data set. (a) RX (AUC = 0.9538). (b) LRX (AUC = 0.9391). (c) CRD (AUC = 0.9450). (d) BJSRD (AUC = 0.9133). (e) LSMAD (AUC = 0.9407). (f) GTD (AUC = 0.9676).

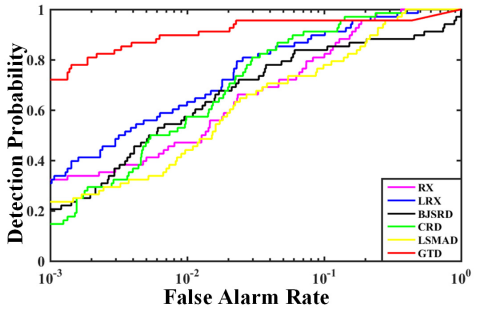


Fig. 10. ROC curves for the Pavia City data set.

fail to remove interferences of background pixels. The ROC curves of different detectors are shown in Fig. 8. We can obviously find that the ROC curve of the GTD method lies nearer the upper leftmost corner. Furthermore, the GTD method can achieve the highest AUC score, 0.9882, which also demonstrates the advantage of the proposed method.

For the Pavia City data set, the detection maps are visualized in Fig. 9. This shows that the GTD method has a better detection ability with respect to other compared methods. The RX, CRD, and LSMAD methods can identify most of the cars but still miss a part of targets. The LRX and BJSRD methods fail to detect targets from background pixels effectively. In Fig. 10, the ROC curves are plotted to measure detection results qualitatively. The GTD method exhibits higher detection probabilities when FARs vary from 0 to 1. Moreover, it can be observed that the AUC score of the GTD method is 0.9676, which is higher than other detectors.

For the Texas Coast data set, the detection results are illustrated in Fig. 11. The LRX and BJSRD detectors lose a few anomalies. Although the CRD and LSMAD methods perform better than LRX and BJSRD methods, they fail to preserve the shape of some targets. The RX and the proposed GTD methods can acquire excellent detection performance, and the GTD method is closest to the reference map. This detector can

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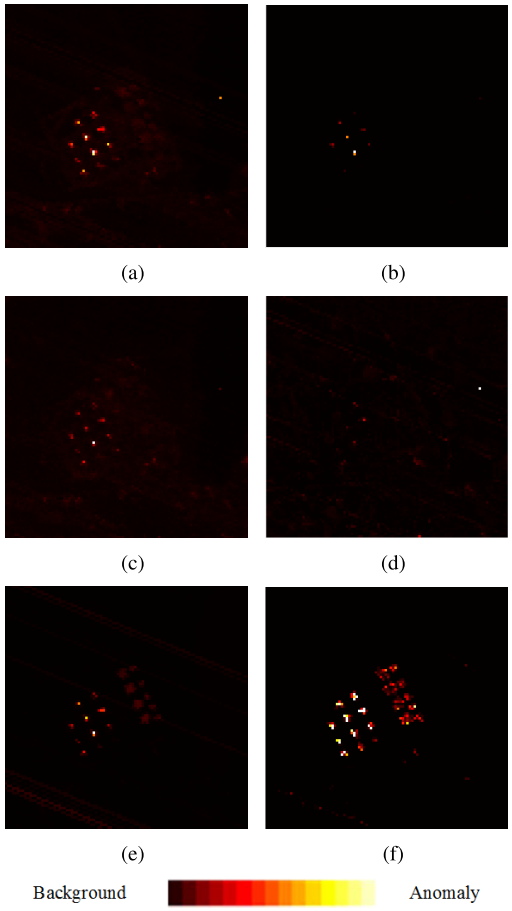


Fig. 11. Detection maps obtained by different methods for the Texas Coast data set. (a) RX (AUC = 0.9946). (b) LRX (AUC = 0.9132). (c) CRD (AUC = 0.9438). (d) BJSRD (AUC = 0.7712). (e) LSMAD (AUC = 0.9854). (f) GTD (AUC = 0.9953).

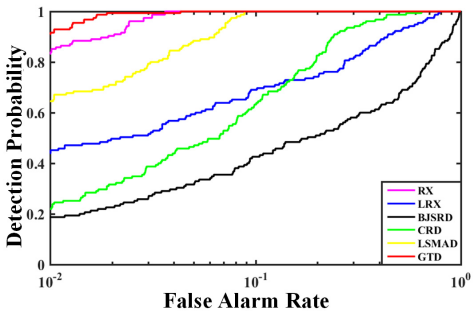


Fig. 12. ROC curves for the Texas Coast data set.

achieve the highest AUC score, 0.9953. The AUC scores of the RX, LRX, CRD, BJSRD, and LSMAD methods are 0.9946, 0.9132, 0.9438, 0.7712, and 0.9854, respectively. In Fig. 12, the ROC curve is shown, which also demonstrates that the proposed method obtains the best detection performance.

For the Los Angeles data set, the detection results are presented in Fig. 13. The RX, CRD, and LSMAD approaches can identify the position of anomaly targets but cannot remove the interference of background pixels effectively. The LRX method misses some anomaly targets. In Fig. 14, the ROC curves are provided, illustrating that the GTD method achieves a higher DP with respect to other methods.

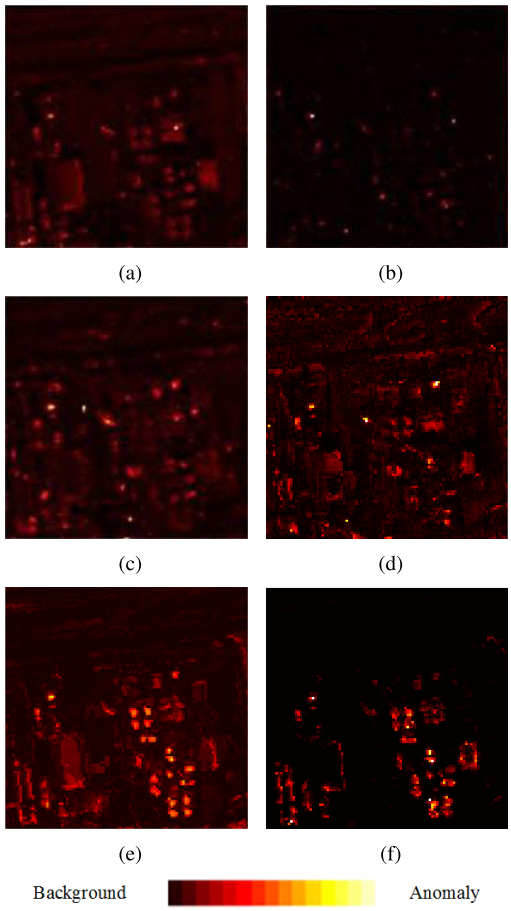


Fig. 13. Detection maps obtained by different methods for the Los Angeles data set. (a) RX (AUC = 0.9692). (b) LRX (AUC = 0.95380). (c) CRD (AUC = 0.9521). (d) BJSRD (AUC = 0.9253). (e) LSMAD (AUC = 0.9607). (f) GTD (AUC = 0.9759).

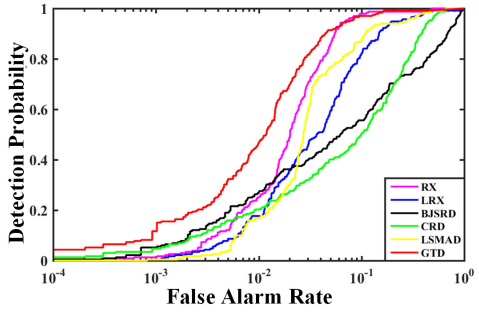


Fig. 14. ROC curves for the Los Angeles data set.

The AUC scores for four data sets are illustrated in Table I together. All methods are implemented on a laptop with 2.6-GHz CPU and 32-G memory, and the test platform is MATLAB R2017a. The running time of the compared methods is presented in Table II. Since extractions of three features in the anomaly game step are independent, a parallel way is used to accelerate the detection processing. It can be seen that the GTD detector is not the most efficient method, but it is much faster than the LRX, BJSRD, and CRD methods.

D. Analysis the Influence of Parameters

In this section, we analyze the influence of two main parameters in the GTD method, i.e., Gaussian weight parameter σ

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TABLE I
AUC SCORE FOR ALL DETECTION RESULTS

Data set	RX [24]	LRX [24]	CRD [40]	BJSRD [36]	LSMAD [43]	GTD
San Diego	0.9403	0.9457	0.9514	0.9643	0.9788	**0.9882**
Pavia City	0.9538	0.9391	0.9450	0.9133	0.9407	**0.9676**
Texas Coast	0.9946	0.9132	0.9438	0.7712	0.9854	**0.9953**
Los Angeles	0.9692	0.9538	0.9521	0.9253	0.9607	**0.9759**

TABLE II
COMPUTING TIME (SECONDS) OF DIFFERENT METHODS

Data set	RX [24]	LRX [24]	CRD [40]	BJSRD [36]	LSMAD [43]	GTD
San Diego	0.28	76.23	81.76	74.53	0.82	7.75
Pavia City	0.47	95.23	136.32	145.53	1.34	10.83
Texas Coast	0.34	81.27	98.83	85.77	0.93	7.92
Los Angeles	0.31	86.97	112.97	103.54	0.84	7.62

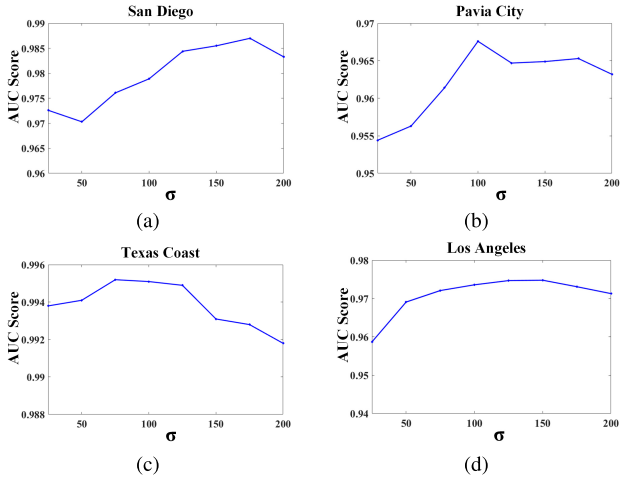


Fig. 15. Effect of the Gaussian weight parameter σ on the detection performance of the proposed GTD method. (a) San Diego. (b) Pavia City. (c) Texas Coast. (d) Los Angeles.

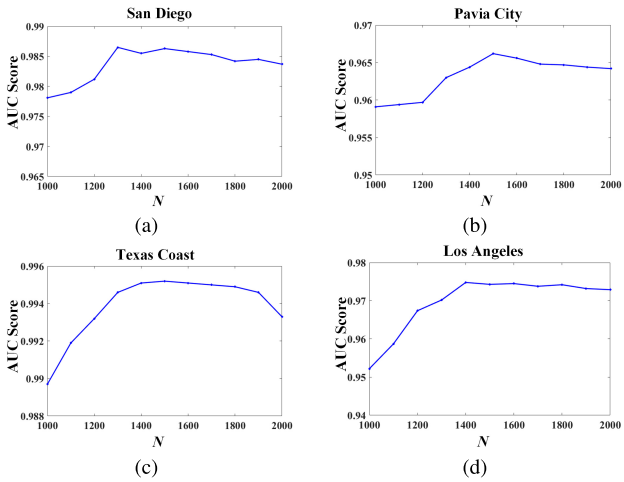


Fig. 16. Effect of the superpixels number N on the detection performance of the proposed GTD method. (a) San Diego. (b) Pavia City. (c) Texas Coast. (d) Los Angeles.

and superpixel number N. Here, the AUC scores are used to evaluate the detection performance of the GTD approach. The influence of the weight parameter σ is shown in Fig. 15. The range of σ in the experiments is $[25, 50, 75, \ldots, 200]$. It can be observed that when $75 \leq \sigma \leq 150$, the proposed method can obtain excellent detection performance on four HSIs. The AUC score will descend if σ is quite small or quite large. The graph weight value $e^{-\sigma||\bar{s_i}-\bar{s_j}||}$ for adjoining superpixels closes to be one if σ is too small. Otherwise, the weight will tend to be zero if σ is quite large. That is, the weight $e^{-\sigma||\bar{s_i}-\bar{s_j}||}$ fails to characterize the affinity among adjoining superpixels effectively in two situations. Fig. 16 analyzes the performance of the proposed method when the superpixels number N is set to $[1000, 1100, \ldots, 2000]$. We can observe that there is a dramatically upward tendency when the superpixels number initially increases from 1000 to 1500. The changing trend for four images is different when N is increased after 1500. For the Los Angeles data set, the relatively stable and high detection accuracy can be acquired when $N \in [1500, 2000]$. For the San Diego, Pavia City, and Texas Coast data sets, there is a slight decrease in the curves, when N is larger than 1500.

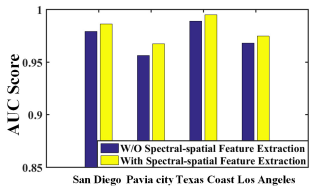


Fig. 17. AUC scores of GTD with and without the spectral–spatial feature extraction step.

The reason is that there exist some homogeneous regions of large size. Thus, if N is quite large, the size of each superpixel will be quite small. San Diego, Pavia City, and Texas Coast data sets have many homogenous regions. That is, the spatial information for these regions fails to be fully exploited.

E. Analysis the Influence of Different Components

In this section, we first analyze the influence of three major steps in the GTD method, including spectral–spatial feature, including spectral–spatial feature extraction, anomaly game, and saliency-based decision fusion. To illustrate the effectiveness of the feature extraction step, we compare the

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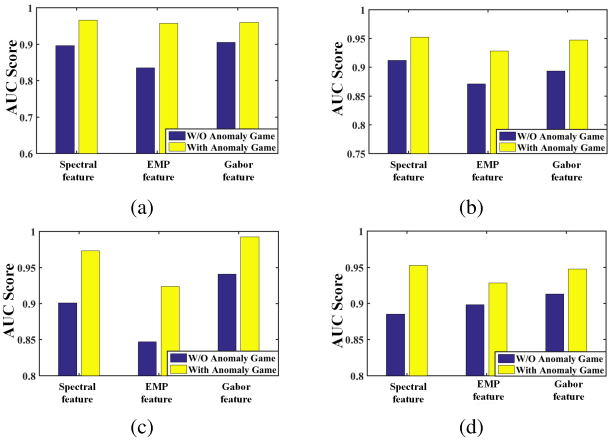


Fig. 18. AUC scores of GTD with and without the anomaly game step. (a) San Diego. (b) Pavia City. (c) Texas Coast. (d) Los Angeles.

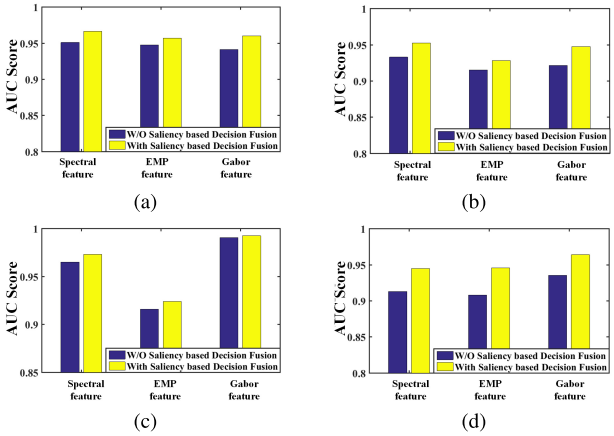


Fig. 19. AUC scores of GTD with and without the saliency-based decision fusion step. (a) San Diego. (b) Pavia City. (c) Texas Coast. (d) Los Angeles.

detection results of the proposed method with or without the spectral–spatial feature extraction step (using the original HSI as the input). Fig. 17 shows the effectiveness of the spectral–spatial feature extraction step on four data sets. It can be observed that the spectral–spatial feature extraction step is very effective. For example, for the Pavia city data set, the AUC score of the GTD method has been improved from 0.9558 to 0.9676. Fig. 18 shows the detection performance of the proposed method with or without the anomaly game step. It can be found that the anomaly game step is instrumental in detecting anomalies. For different features, the AUC scores all have been greatly improved. Furthermore, we explore the effect of saliency-based decision fusion. For different features, the AUC scores obtained by the proposed method with or without this step are shown in Fig. 19. It is observed that the saliency-based decision fusion plays an important role in improving the detection performance.

Then, we demonstrate the advantage of the adopted superpixels segmentation method and the adopted saliency weight construction method. To illustrate the effectiveness of the ERS method, a simple linear iterative clustering (SLIC) method [64] also has been employed to generate superpixels. In Fig. 20, the detection results obtained by the two superpixel

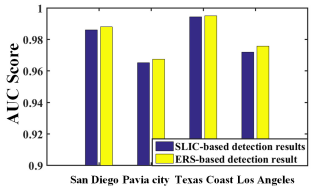


Fig. 20. Detection results obtained by two superpixel segmentation methods.

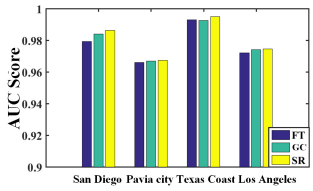


Fig. 21. Evaluation scores of three methods on four data sets.

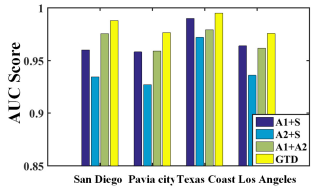


Fig. 22. Detection results of each term in the payoff function on four data sets.

segmentation methods have been shown. The number of superpixels is the same in the anomaly game. The first three principal components are used as the input in the SLIC method. It can be observed that detection results obtained by the two superpixel segmentation methods have outstanding detection performance, and the ERS method can lead to a better detection result. Therefore, the ERS method is adopted for segmentation in this article. To illustrate the effectiveness of the SR method [56], we compare the SR model with two famous saliency detection methods, global contrast (GC)-based method [65] and frequency-tuned (FT)-based method [66] to estimate the weight maps W_n for three features. In Fig. 21, the detection results of the compared methods have been shown. We can observe that the detection results of the compared methods both have outstanding detection performance, and the SR method has more competitive detection performance.

Moreover, we analyze the influence of each term in (1). We estimate the detection result when one of them is lost. Fig. 22 shows the detection results in each case. $A1$ and $A2$ represent two anomaly terms, $ano^1(i)$ and $ano^2(i)$. S represents the support term, $\sup_{ij}(z_i, z_j)$. It can be observed that each component has its distinctive influence on detection results.

Last, we analyze the influence of each spectral–spatial feature. The detection results of the spectral feature can benefit from the spectral information in HSIs. The EMP feature can reflect the spatial morphology property in HSIs. The Gabor feature can exploit the spatial texture information in HSIs for detection. In Fig. 23, we show the detection maps of three features and the GTD method on the San Diego data set. It can be observed that map_1 obtains the best detection performance

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TABLE III
INFLUENCE OF THREE FEATURES ON FOUR DATA SETS

Data set	S	E	G	S+E	S+G	E+G	GTD
San Diego	0.9667	0.9572	0.9604	0.9755	0.9842	0.9746	**0.9882**
Pavia City	0.9529	0.9286	0.9478	0.9586	0.9637	0.9519	**0.9676**
Texas Coast	0.9734	0.9241	0.9927	0.9767	0.9945	0.9939	**0.9953**
Los Angeles	0.9529	0.9286	0.9478	0.9584	0.9707	0.9683	**0.9759**

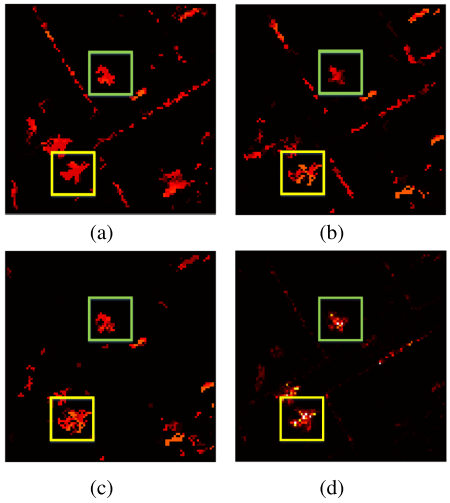


Fig. 23. Detection maps of three features and the GTD method. (a) Detection map_1 generated by our anomaly game in spectral feature. (b) Detection map_2 generated by our anomaly game in EMP feature. (c) Detection map_3 generated by our anomaly game in Gabor texture feature. (d) Final detection map.

in the yellow region, whereas map_2 and map_3 illustrate that the shape of the target cannot be kept effectively. A similar situation can be found in the green region. map_2 has the best detection performance in this region. Meanwhile, map_3 has the best performance in the suppression of background pixels. There is strong complementary information in three features. The reason is that they are extracted from the same HSI from different aspects. By jointly utilizing three features, the GTD method can achieve a better detection performance. Moreover, Table III illustrates the influence of these features on four data sets. "S," "E," and "G" represent spectral, EMP, and Gabor features, respectively. It can be found that each feature has a distinctive contribution to the final detection result.

V. CONCLUSION

This article has proposed a novel game theory-based approach for hyperspectral anomaly detection by exploring multiple spectral and spatial characteristics of anomalies. First, with a new strategic interaction perspective, the proposed method defines the anomaly detection problem as an anomaly game model. Detection maps can be generated effectively based on each player's strategy in the Nash equilibrium of the anomaly game. Then, a saliency-based decision fusion technique is used to combine the complementary information in different features, so as to obtain a fused detection map. Experiments performed on four real HSIs demonstrate that our detector outperforms its counterparts in terms of detection accuracy.

The proposed method could still be further applied in other remote sensing fields (such as scene classification and mountain fire detection). Moreover, most of the detection methods including the proposed method are designed for HSIs of small-scale. Thus, improving the proposed method for anomaly detection in HSIs of large-scale is a challenging but interesting future work.

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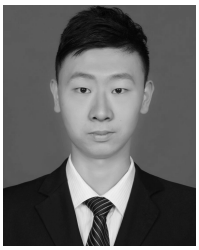
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